SWINBURNE UNIVERSITY OF TECHNOLOGY

# DEPLOYMENT PORTFOLIO TASK 1

My Pass to Distinction Implementation and Evidence Report

**Student:** Sidharth Krishnan
**Student ID:** 104089870
**Unit:** SWE40006 Software Deployment and Evolution
**Teaching period:** Semester 2, 2026
**Submission date:** 14 August 2026
**Declared target:** Task 1.3 - Distinction
**Source repository:** https://github.com/RealSid08/SWE40006_Task1

**Submission statement:** I completed Tasks 1.1, 1.2, and 1.3 on Windows. I implemented the applications and WiX projects, built both MSI packages, and verified the required installation, execution, and uninstallation for Task 1.1. I then reinstalled both final packages so the applications remain available on my PC. For Distinction, I explicitly packaged three external runtime DLLs: Newtonsoft.Json, CsvHelper, and Humanizer.

# 1. Executive summary

In this portfolio, I completed an end-to-end Windows deployment workflow with WiX Toolset 4.0.6 and .NET 10. For Task 1.1, I packaged a Hello World console application. For Task 1.2, I designed and built HashGuard Desktop, a WinForms checksum utility. For Task 1.3, I extended the HashGuard installer to package and verify multiple external DLL dependencies. Both final MSI packages are installed on my PC, and their deployed application files are present.

> **Outcome:** I ran the final automated verification and all 4 tests passed. I rebuilt both MSI projects with 0 warnings and 0 errors. The final Hello World and HashGuard installations completed successfully, and I confirmed that their installed files are present. HashGuard is also available from its Start Menu shortcut.

## 1.1 Setup environment and toolchain

| | |
|---|---|
| **Host environment** | Native Windows 11 Home, 64-bit, build 26200 |
| **VM details** | Not applicable - I completed the assignment directly on a Windows PC, not a macOS virtual machine |
| **Solution** | SWE40006.Task1.sln containing five application, test, and installer projects |
| **Build target** | Release configuration; win-x64 application publish and x64 MSI packages |

| Component | Installed version | Role |
|---|---|---|
| Visual Studio | Community 2026 18.9.0 | .NET desktop workload and solution/project authoring |
| .NET | SDK 10.0.400; Desktop Runtime 10.0.11 | Build, test, publish, and execute the applications |
| WiX Toolset | CLI and SDK 4.0.6 | Compile the two x64 MSI packages |
| HeatWave | Visual Studio extension installed | WiX project templates and Visual Studio integration |

## 1.2 Visual Studio solution setup

I opened the complete SWE40006.Task1 solution in Visual Studio and confirmed that it contained both WiX installer projects plus the Hello World, HashGuard Desktop, and HashGuard test projects. This kept the application, tests, and deployment source together while still allowing each project to be built independently.

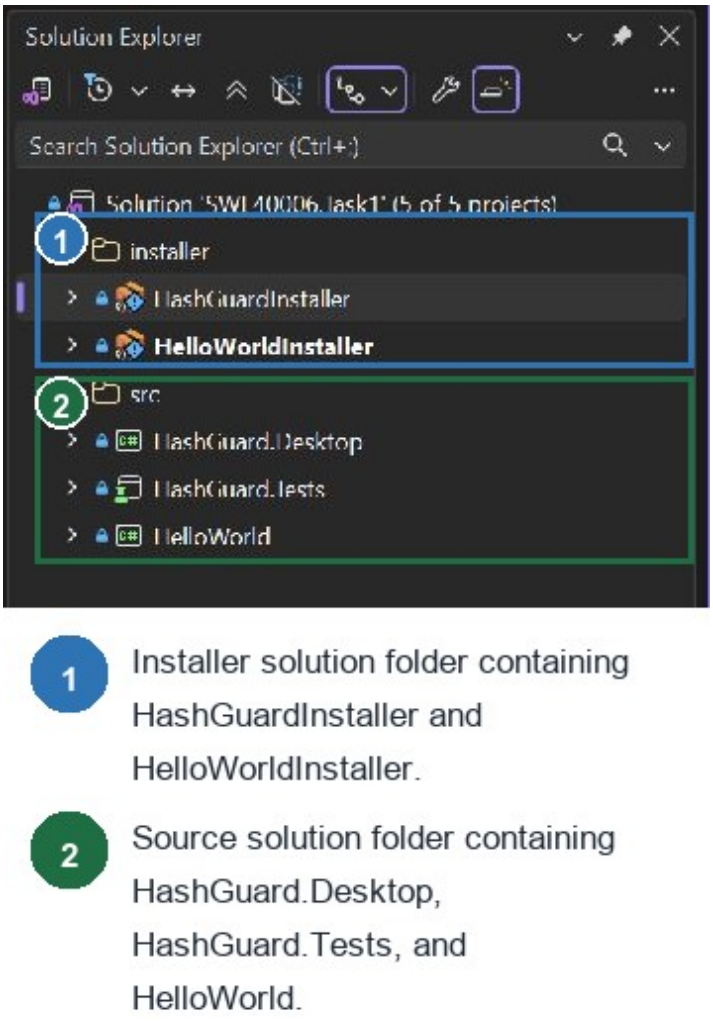



*Figure 1. Annotated Visual Studio Solution Explorer showing the two WiX installer projects and the three application/test projects.*

## 1.3 Step-by-step setup and build workflow

1. **Set up the Windows toolchain:** I used Visual Studio Community 2026 18.9.0 with the .NET desktop development tools, .NET SDK 10.0.400, .NET Desktop Runtime 10.0.11, WiX Toolset 4.0.6, and the HeatWave Visual Studio extension.
2. **Open and verify the solution:** I opened SWE40006.Task1.sln and checked that Visual Studio loaded HelloWorld, HashGuard.Desktop, HashGuard.Tests, HelloWorldInstaller, and HashGuardInstaller.
3. **Restore, build, and test the applications:** I restored the pinned NuGet packages, built the C# projects in Release mode, and ran the HashGuard test project. The final test run passed all four tests with no failures or skipped tests.
4. **Publish the deployable application files:** I published Hello World and HashGuard for win-x64 as framework-dependent applications so each installer had a stable Release payload containing the executable, application DLL, dependency manifest, and runtime configuration.

5. **Author the WiX packages:** I defined the installation directories, application files, product identity, upgrade behaviour, and shortcuts in the .wxs files. For Task 1.3, I also authored separate WiX components for Newtonsoft.Json.dll, CsvHelper.dll, and Humanizer.dll.
6. **Build both MSI packages:** I compiled HelloWorldInstaller and HashGuardInstaller in Release mode. Both final builds produced their expected x64 MSI file with 0 warnings and 0 errors.
7. **Install and inspect the deployed files:** I installed each MSI with Windows Installer verbose logging, checked the exit status, and inspected the installed directories. Both installation logs ended with status 0, and the expected files were present.
8. **Execute and verify the installed applications:** I ran Hello World from its installed directory and captured its banner. I launched the installed HashGuard application, calculated both checksums for the sample payload, and confirmed that a history record was saved. I also completed the Task 1.1 uninstall check and reinstalled the final packages.

### 1.4 Console proof of successful builds and installs

**Final Release build output - evidence/31 and evidence/32**

```
HelloWorldInstaller -> ...\SWE40006-HelloWorld-1.0.0-x64.msi
Build succeeded.
    0 Warning(s)
    0 Error(s)

HashGuardInstaller -> ...\HashGuard-Desktop-1.0.0-x64.msi
Build succeeded.
    0 Warning(s)
    0 Error(s)
```

**Final Windows Installer log excerpts - evidence/43 and evidence/44**

```
Windows Installer installed the product. Product Name: SWE40006 Hello World.
Installation success or error status: 0.
MainEngineThread is returning 0

Windows Installer installed the product. Product Name: HashGuard Desktop.
Installation success or error status: 0.
MainEngineThread is returning 0
```

### 1.5 Reproducibility controls

I pinned the WiX SDK and extension versions in the .wixproj files and locked the NuGet dependency versions. I also used stable product and upgrade GUIDs. I retained my final test, build, installation, execution, and metadata logs in the evidence directory, and I recorded both MSI SHA-256 hashes in evidence/33-msi-metadata.txt.

## 2. Task 1.1 - Pass: Hello World MSI

For the Pass task, I created a small .NET console program that prints an assignment-specific banner. I then authored a WiX x64 package that installs the published executable and its three companion runtime files to %LOCALAPPDATA%\Programs\SWE40006 Hello World. I installed the MSI silently, ran the installed executable from that directory, and captured the output. I also verified that the package could be uninstalled as required, then reinstalled the final MSI so Hello World remains on my PC.

| | |
|---|---|
| **MSI** | SWE40006-HelloWorld-1.0.0-x64.msi |
| **Product version** | 1.0.0 |
| **Product code** | {40C76FDB-606D-4DE6-9B17-F06149341B0A} |
| **Install result** | Exit code 0; four application files present |
| **Execution result** | Installed application printed the expected Task 1.1 banner |
| **Final state** | Package reinstalled; four application files remain present |

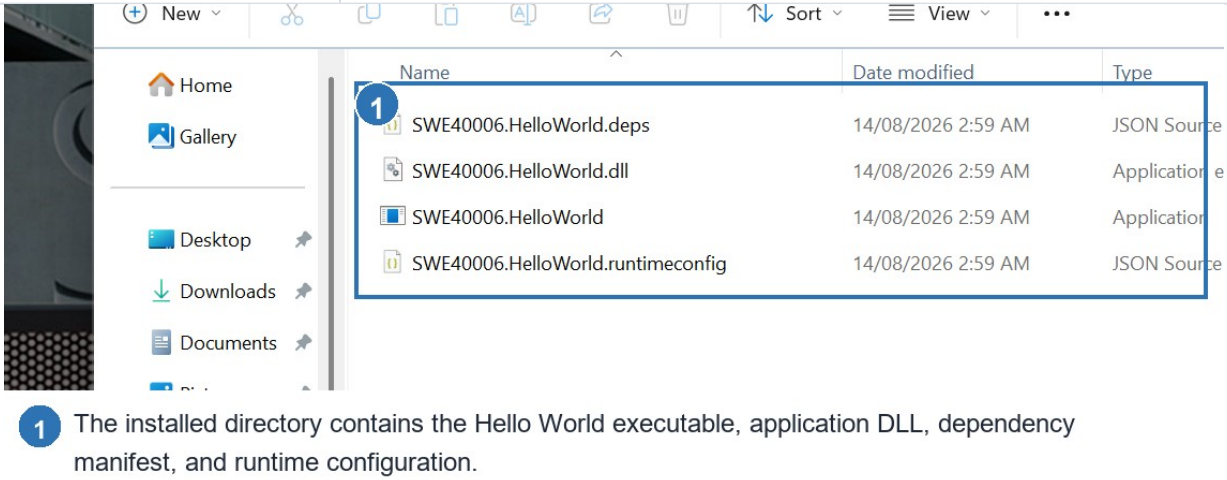



*Figure 2. Annotated installed-directory evidence for the Task 1.1 Hello World package.*

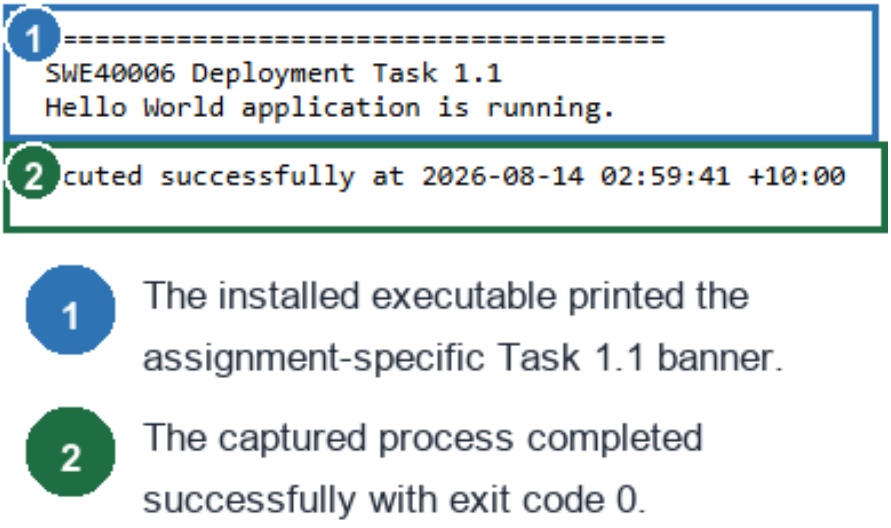


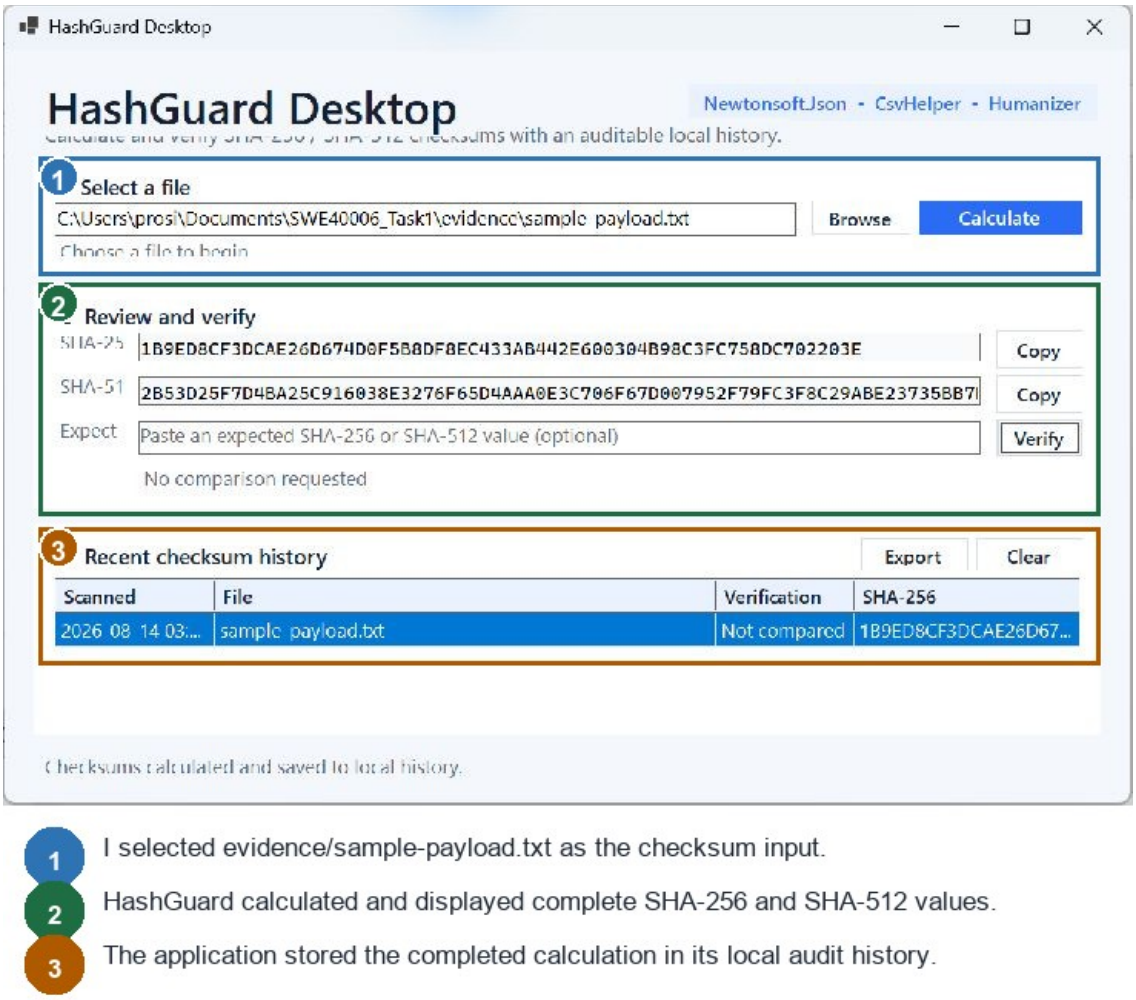

*Figure 3. Annotated console output from the installed Hello World executable.*

# 3. Task 1.2 - Credit: HashGuard Desktop

For the Credit task, I designed and built HashGuard Desktop as an original C# WinForms application for calculating SHA-256 and SHA-512 file checksums. I implemented asynchronous hashing, optional comparison with an expected hash, a local JSON audit history, and CSV export. I also made the external dependencies visible in the interface and displayed the complete hash values for copying and verification.

## 3.1 Application design

| Layer | Implementation | Responsibility |
|---|---|---|
| Presentation | MainForm.cs | File selection, progress, hash display, comparison, history, and export actions |
| Hashing | HashService.cs | Buffered asynchronous SHA-256/SHA-512 computation using IncrementalHash |
| Persistence | HistoryStore.cs | Auditable local JSON history using Newtonsoft.Json |
| Export | CsvExportService.cs | CSV output through CsvHelper |
| Presentation helper | Humanizer | Readable dates and display text |




1. I selected evidence/sample-payload.txt as the checksum input.
2. HashGuard calculated and displayed complete SHA-256 and SHA-512 values.
3. The application stored the completed calculation in its local audit history.

***Figure 4. Annotated execution evidence from the installed HashGuard application.***

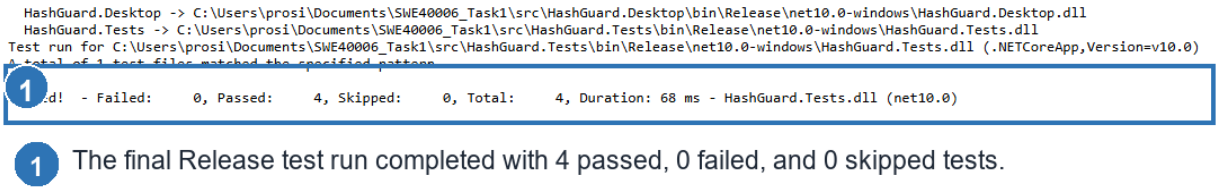
## 3.2 Core checksum implementation

**C# excerpt - HashService.ComputeAsync**

```
        using var sha256 = IncrementalHash.CreateHash(HashAlgorithmName.SHA256);
        using var sha512 = IncrementalHash.CreateHash(HashAlgorithmName.SHA512);
        await using var stream = new FileStream(
            filePath,
            FileMode.Open,
            FileAccess.Read,
            FileShare.Read,
            BufferSize,
            FileOptions.Asynchronous | FileOptions.SequentialScan);

        var buffer = new byte[BufferSize];
        long totalRead = 0;
        int bytesRead;

        while ((bytesRead = await stream.ReadAsync(buffer, cancellationToken)) > 0)
        {
            sha256.AppendData(buffer, 0, bytesRead);
            sha512.AppendData(buffer, 0, bytesRead);
            totalRead += bytesRead;
            progress?.Report(fileInfo.Length == 0 ? 100 : (int)(totalRead * 100 / fileInfo.Length));
        }

        progress?.Report(100);
        return new HashResult(
            fileInfo.FullName,
            fileInfo.Length,
            Convert.ToHexString(sha256.GetHashAndReset()),
            Convert.ToHexString(sha512.GetHashAndReset()));
```

I implemented HashService to stream each file in 1 MiB buffers and append every buffer to both hash algorithms. This avoids loading the entire file into memory and allows the interface to display progress for large inputs.

## 3.3 Automated verification




1 The final Release test run completed with 4 passed, 0 failed, and 0 skipped tests.

*Figure 5. Annotated console output from the final automated test run.*

| Test | Expected behaviour | Result |
|---|---|---|
| Known hashes | SHA-256 and SHA-512 match known values | Passed |
| Mismatch verification | Malformed or different expected hash is rejected | Passed |
| JSON round trip | Newtonsoft.Json persists and restores a history record | Passed |
| CSV export | CsvHelper writes an export containing the history data | Passed |

# 4. Task 1.3 - Distinction: external dependencies

**Distinction criterion:** To satisfy the Distinction criterion, I authored multiple runtime dependencies as individual WiX components. I verified that the final HashGuard installation contains Newtonsoft.Json.dll, CsvHelper.dll, and Humanizer.dll alongside the application.

## 4.1 Runtime dependency set

| Dependency | Pinned version | Use in HashGuard |
|---|---|---|
| Newtonsoft.Json | 13.0.4 | Serializes and deserializes checksum history |
| CsvHelper | 33.1.0 | Exports the audit history to CSV |
| Humanizer.Core | 3.0.10 | Formats readable time and display values |

## 4.2 WiX authoring

**WiX v4 excerpt - explicit external DLL components**

```
<Component Id="NewtonsoftJsonComponent" Guid="{9AAAABB3-2199-4AFB-B3F9-14AAAB88A66A}">
<File Id="NewtonsoftJsonDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
x64\publish\Newtonsoft.Json.dll" KeyPath="yes" />
<Component Id="CsvHelperComponent" Guid="{F6EBD07F-2291-4527-9317-4739766DD50B}">
<File Id="CsvHelperDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
x64\publish\CsvHelper.dll" KeyPath="yes" />
<Component Id="HumanizerComponent" Guid="{EA9E4267-5AA4-4E47-9AAA-463335F372FD}">
<File Id="HumanizerDll" Source="..\..\src\HashGuard.Desktop\bin\Release\net10.0-windows\win-
x64\publish\Humanizer.dll" KeyPath="yes" />
```

I assigned every external file its own stable component GUID and KeyPath. In the same package, I defined the application executable, application DLL, dependency manifest, runtime configuration, installation directory, and Start Menu shortcut.

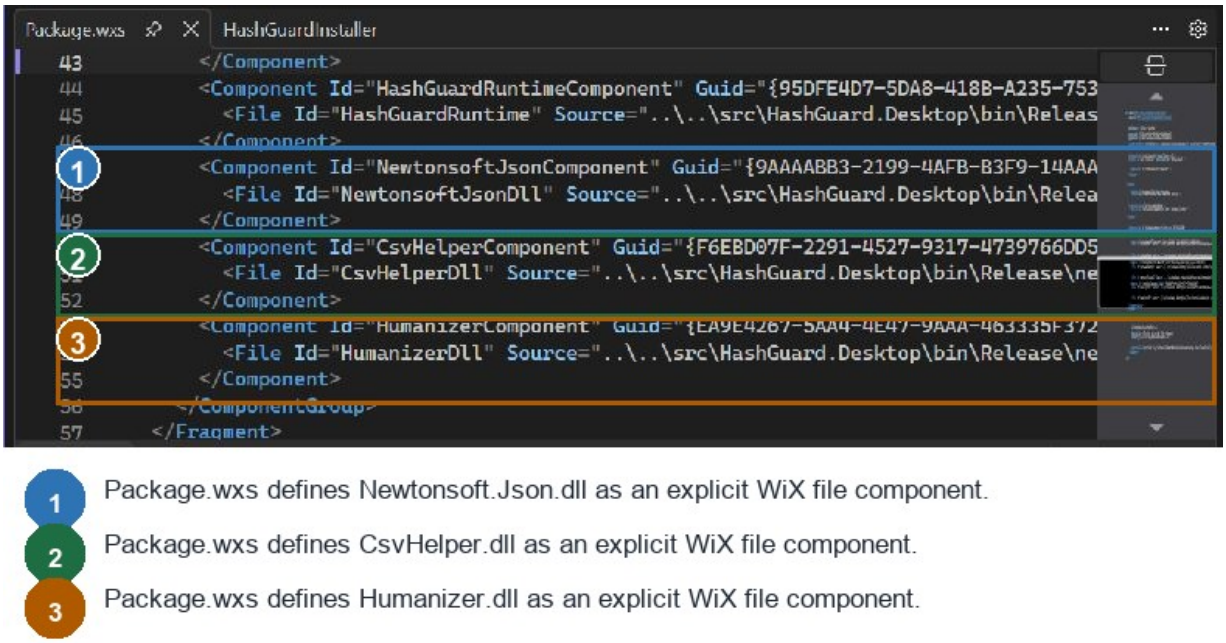



*Figure 6. Annotated Visual Studio view of my Package.wxs external-DLL components.*

## 4.3 Build and installed-file evidence

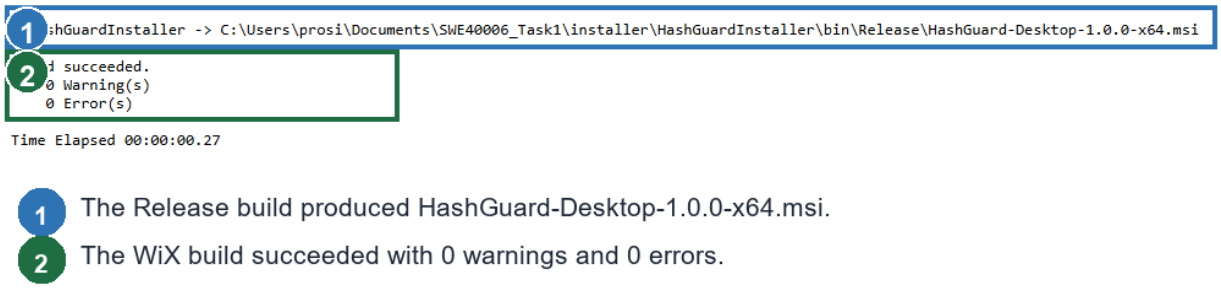



*Figure 7. Annotated console output from the final HashGuard MSI build.*

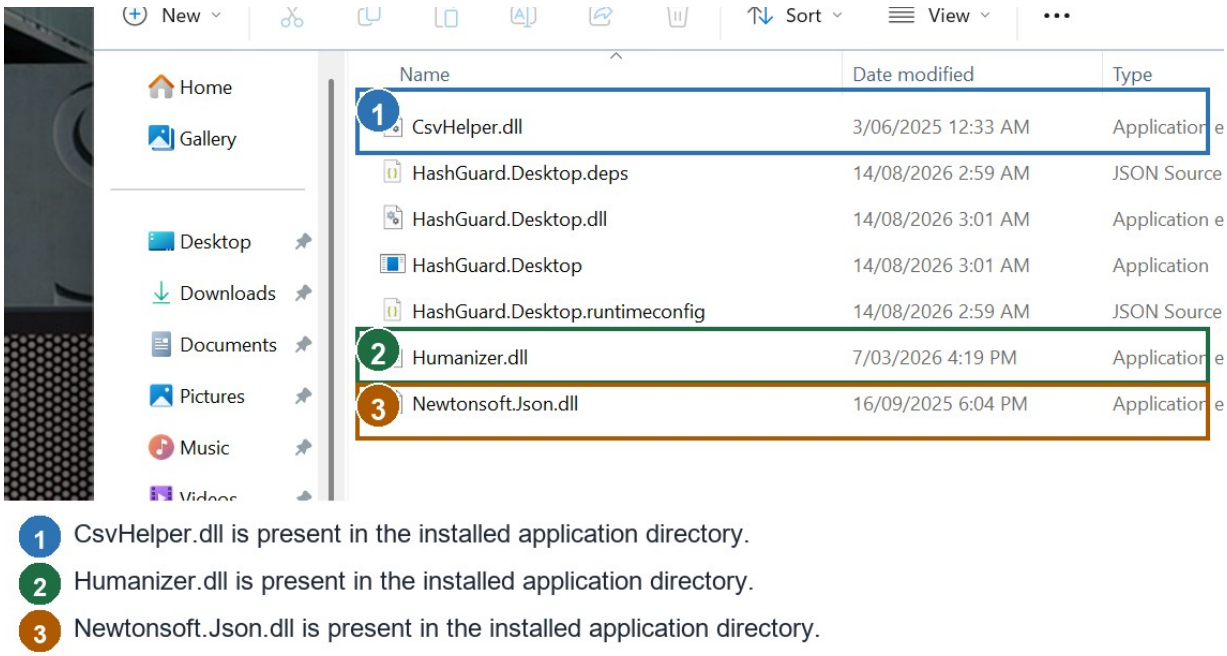



*Figure 8. Annotated installed-directory evidence for the three external runtime DLLs.*

## 4.4 MSI identity and installation

| | |
|---|---|
| **MSI** | HashGuard-Desktop-1.0.0-x64.msi |
| **Product code** | {C57B16A8-E9E5-4EE9-8020-EE069EB409A7} |
| **Upgrade code** | {23AB1A31-A44A-41FE-BBDA-5245030BAD76} |
| **Size** | 1,093,632 bytes |
| **SHA-256** | 8B44ED575B560DD0C996AEB5CC1CF06C06DE8C9A3F31D669A3C50134AB488ED1 |
| **Final install** | Exit code 0; 7 files plus Start Menu shortcut |
| **Execution** | Installed application launched and processed evidence/sample-payload.txt |

# 5. Verification and evidence matrix

| Requirement | Evidence | Verified result |
|---|---|---|
| Visual Studio setup | 47-visual-studio-solution-and-wix.png | Five projects loaded in Visual Studio; Package.wxs opened for inspection |
| Toolchain | 27-toolchain.txt, 28-dotnet-tools.txt, 29-visual-studio-workload.json | .NET, WiX, HeatWave/Visual Studio workload recorded |
| Pass build | 31-final-hello-msi-build.log | 0 warnings; 0 errors |
| Pass lifecycle | 06-hello-install.log, 08-hello-execution.log, 41-hello-uninstall.log, 43-restored-hello-install.log | Required installation, execution, and uninstallation verified; final package reinstalled |
| Credit tests | 30-final-tests.log | 4 passed; 0 failed |
| Distinction build | 32-final-hashguard-msi-build.log | 0 warnings; 0 errors |
| Distinction install | 44-restored-hashguard-install.log, 20-hashguard-final-files.txt | Exit code 0; 3 external DLLs present in final installation |
| MSI metadata | 33-msi-metadata.txt | Product identity and SHA-256 hashes recorded |

## 5.1 Manual checksum result

| | |
|---|---|
| **Input** | evidence/sample-payload.txt |
| **SHA-256** | 1B9ED8CF3DCAE26D674D0F5B8DF8EC433AB442E600304B98C3FC758DC702203E |
| **SHA-512** | 2B53D25F7D4BA25C916038E3276F65D4AAA0E3C706F67D007952F79FC3F8C29AB E23735BB7D6E1A41C3AE4A89D9EE1694A82E1894C4A9EBC3F1B121A640C47AF |
| **History** | One record written to the local JSON audit history |

# 6. Troubleshooting and engineering decisions

## 6.1 WinForms designer serialization warning

During the initial compilation, I encountered WFO1000 because WinForms designer serialization interpreted public form state as design-time content. I refactored the form so its application services and state are private readonly fields initialized in the constructor. This separated runtime state from designer serialization and restored a clean build.

## 6.2 WiX v4 UI migration

My initial WiX build used the WiX v3 UIRef pattern and failed with WIX0094 because that identifier is protected in WiX v4. I migrated the package to the WiX v4 UI extension namespace and declared ui:WixUI with WixUI_InstallDir and INSTALLFOLDER. I pinned WixToolset.UI.wixext to version 4.0.6.

## 6.3 Installed-app runtime failure

When I installed my first HashGuard build, it terminated before displaying its window. I checked the Windows Application Event Log and identified a .NET 10 NotSupportedException caused by a transparent button border setting. I corrected the custom styling by explicitly setting BorderSize to 0, then repeated the tests, publish, MSI build, installation, and installed-application execution successfully.

## 6.4 MSI validation policy

On my first non-elevated WiX build, ICE validation could not run under the available Windows policy. I compensated by inspecting Windows Installer COM metadata, calculating SHA-256 hashes, performing real silent installations, checking installed files, running each application from its deployed directory, reviewing MSI logs, and confirming successful execution. My final reproducibility builds completed with 0 warnings and 0 errors.

## 6.5 Packaging decisions

| Decision | Reason | Effect |
|---|---|---|
| Per-user MSI | Avoids unnecessary elevation for a student utility | Installs under LocalAppData and supports silent testing |
| Explicit DLL components | Makes Distinction dependency evidence unambiguous | Each external DLL has a stable component GUID and key path |
| Framework-dependent publish | Keeps the MSI small while targeting the installed Desktop Runtime | Runtimeconfig and deps files are deployed with the application |

# 7. Reproduction procedure

I used the following commands from the repository root on Windows with .NET 10 and WiX Toolset 4.0.6 installed. Because I pinned the project dependencies, these commands can be rerun to reproduce my test, publish, build, and installation results.

**PowerShell - build and test**

```
dotnet test src/HashGuard.Tests/HashGuard.Tests.csproj -c Release
dotnet publish src/HashGuard.Desktop/HashGuard.Desktop.csproj -c Release -r win-x64 --self-contained
false
dotnet build installer/HelloWorldInstaller/HelloWorldInstaller.wixproj -c Release
dotnet build installer/HashGuardInstaller/HashGuardInstaller.wixproj -c Release
```

**PowerShell - install**

```
msiexec /i installer\HelloWorldInstaller\bin\Release\SWE40006-HelloWorld-1.0.0-x64.msi /qn /L*v
evidence\43-restored-hello-install.log
msiexec /i installer\HashGuardInstaller\bin\Release\HashGuard-Desktop-1.0.0-x64.msi /qn /L*v
evidence\44-restored-hashguard-install.log
```

## 7.1 Expected outputs

| Step | Success condition | Final observed result |
|---|---|---|
| Test | All automated tests pass | 4/4 passed |
| Build | Both MSI projects succeed | 0 warnings; 0 errors |
| Install | MSI exits 0 and files appear | Both exits 0; 4 and 7 files respectively |
| Run | Installed executable starts | Hello banner and HashGuard checksum result captured |

# 8. References

FireGiant. HeatWave Community Edition and Visual Studio integration. https://docs.firegiant.com/heatwave/

FireGiant. WiX Toolset v4 UI extension (WixUI). https://docs.firegiant.com/wix/tools/wixext/wixui/

Microsoft. IncrementalHash class. https://learn.microsoft.com/dotnet/api/system.security.cryptography.incrementalhash

NuGet. Newtonsoft.Json 13.0.4. https://www.nuget.org/packages/Newtonsoft.Json/13.0.4

NuGet. CsvHelper 33.1.0. https://www.nuget.org/packages/CsvHelper/33.1.0

NuGet. Humanizer.Core 3.0.10. https://www.nuget.org/packages/Humanizer.Core/3.0.10